

UiPath RPA

Workflows & Selectors • Orchestrator Deployment • Queues & Transactions

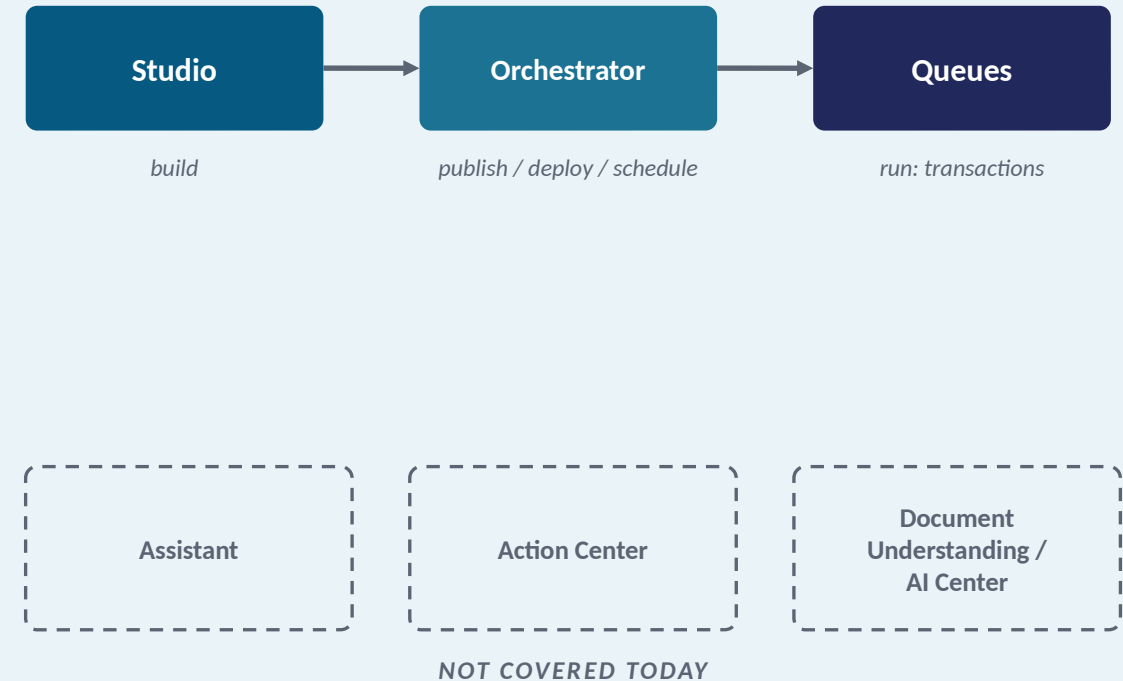


What Is UiPath / RPA?

OVERVIEW

- RPA (Robotic Process Automation): software that operates a screen the way a person would — clicking, typing, reading — instead of calling an API
- A different integration strategy from yesterday's connector-based, structured-data automation
- Studio: where a UiPath workflow is built
- Orchestrator: where it's deployed, scheduled, and monitored
- Queues: transaction-based, retrievable work items processed through a deployed workflow

TODAY: BUILD → DEPLOY → RUN

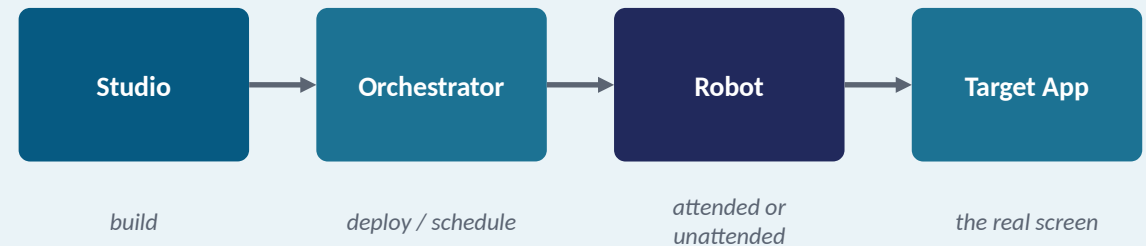


Real, current UiPath product-family members — today's hands-on builds only the top row

A Quick Peek at the Architecture

OVERVIEW

- Studio, Orchestrator, and queues aren't the whole picture — here's how a workflow actually runs end to end
- Studio: where the workflow is built (already covered)
- Orchestrator: publishes, deploys, schedules, and monitors it as a running process
- Robot: the piece that actually executes the workflow — attended (a person triggers it and watches) or unattended (runs alone, on a schedule)
- The robot operates the real application screen — the exact same one a person would use



Module 19A's own selector-based workflow is exactly what runs inside this Robot box once it's deployed

Robot licensing and execution mode are governance topics — out of scope today

UiPath Workflow Basics & Selectors

WHAT'S AHEAD

What RPA automates: a screen, not an API

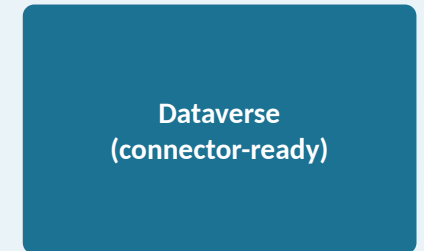
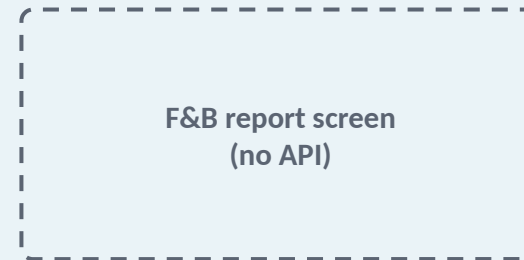
Workflows, activities, and sequences

Selectors: identifying a UI element reliably

No Clean Connector for This Screen

PROBLEM

- Meridian Resorts & Spa must reconcile nightly F&B covers against expected occupancy
- The F&B-covers figure most plausibly lives in a point-of-sale report screen with no API
- Yesterday's Power Automate flow only ever reached structured, connector-exposed data
- Nobody built a door into this screen for anything else to call

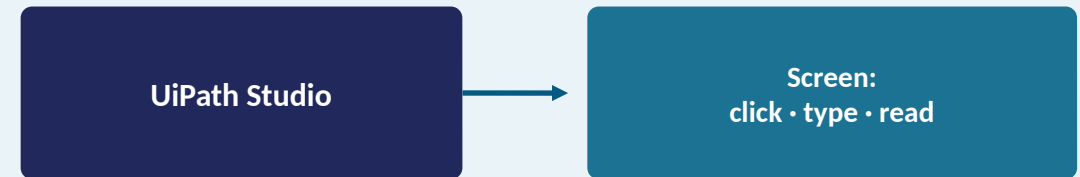


The same comparison yesterday's content named — now the room builds the other side

RPA Automates the Screen, Not the API

SOLUTION

- RPA: software that operates a UI the way a person would — click, type, read
- UiPath Studio is where a workflow is actually built
- A workflow is a file that runs a sequence of automated steps
- Useful precisely where no clean structured-data connector exists



not: calling an API behind the screen

the same screen a person already uses

Meridian Resorts & Spa's real gap — no connector exists, but the screen does

An Empty Workflow Container

PROBLEM

- A new workflow file is an empty container by itself
- Something has to actually go inside it, step by step
- The room's own code had functions and statements; Power Automate had actions
- UiPath needs its own equivalent building block



code: functions · Power Automate: actions · UiPath: ?

Naming Studio didn't yet say what a workflow is made of

Activities and Sequences

SOLUTION

- Activities: individual automated steps, chained one after another
- A Sequence: "suitable to linear processes," runs top to bottom, no branching
- Flowcharts and State Machines exist for decision-heavy automations — not needed today
- Recorder types (Basic, Desktop, Web) can generate activities automatically

SEQUENCE
(top to bottom, no
branching)

Use Application/
Browser

Get Text

Log Message

Flowchart /
State Machine

not built today

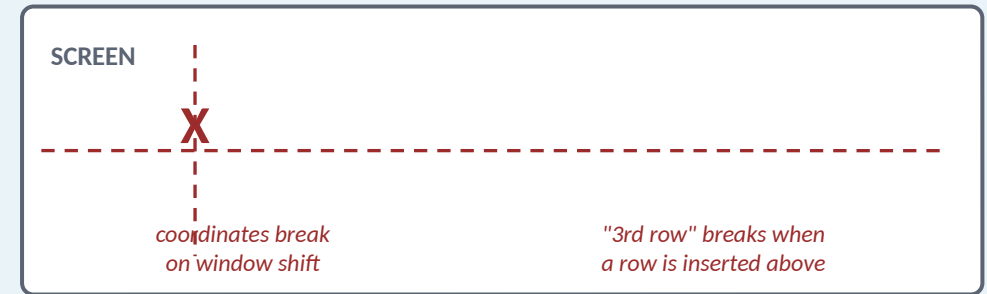
HANDS-ON Build a basic UiPath workflow using selectors to identify UI elements reliably.

Today's own workflow is exactly this shape: one Sequence, three activities

Naming a UI Element Reliably

- An activity has to say precisely which button, label, or value it means
- Screen coordinates break the moment a window shifts by a few pixels
- "The third row" breaks the moment a row gets added above it
- Neither survives the same workflow running more than once

PROBLEM



Neither survives the same workflow running a second time

Selectors Identify an Element Structurally

SOLUTION

- A selector: "an XML fragment that stores the attributes... and its parents"
- Identifies an element by what it structurally is, not where it sits on screen
- Pick attributes with a constant value — a changing one breaks the selector
- Full selector (whole window path) vs. partial selector (inside a container)

HANDS-ON Build a basic UiPath workflow using selectors to identify UI elements reliably.



partial selectors live inside a Use Application/Browser container

A stable attribute chain, not a screen coordinate — this is what survives reruns

Orchestrator & Queues Essentials

WHAT'S AHEAD

Publishing and deploying a workflow to Orchestrator

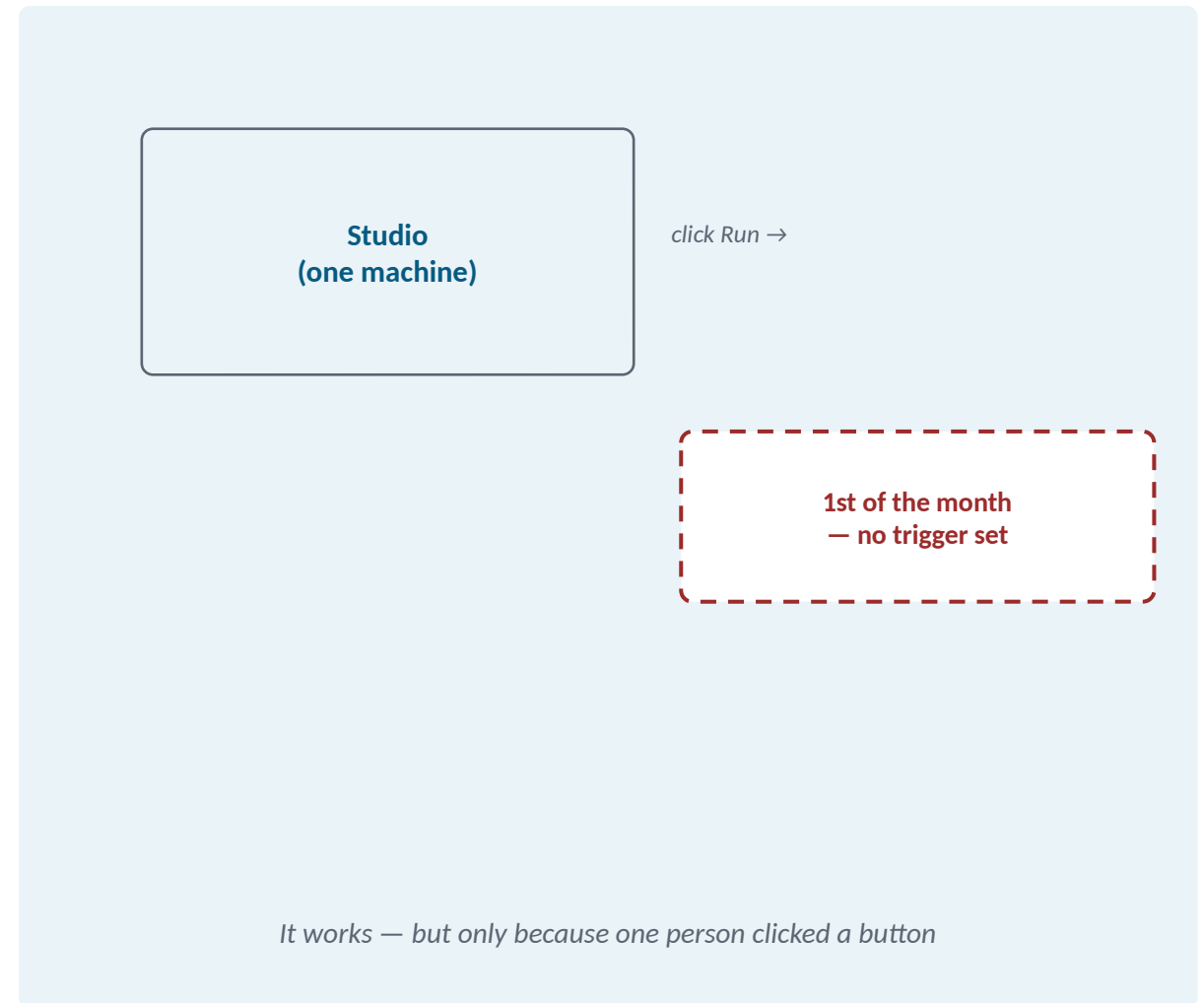
Triggers and scheduling: making a process run on its own

Queues and transaction-based processing, with retries

A Workflow Only One Person Can Run

PROBLEM

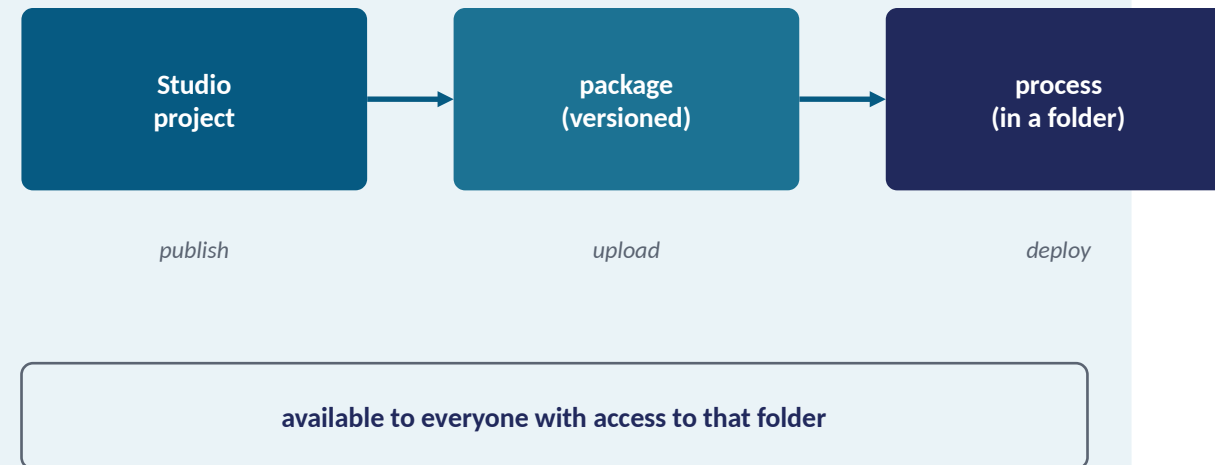
- Module 19A's workflow only runs from inside Studio, on one machine
- Only its own author can start it, whenever they remember to
- Meridian Residences needs its invoice job to run on the 1st of every month
- A workflow nobody but its author can start isn't a dependable automation



Publish and Deploy to Orchestrator

SOLUTION

- Publish: package the Studio project into a versioned artifact
- Upload that package to Orchestrator's package repository
- "A process represents a package version linked to a particular folder"
- Deploying makes it available to everyone with access to that folder



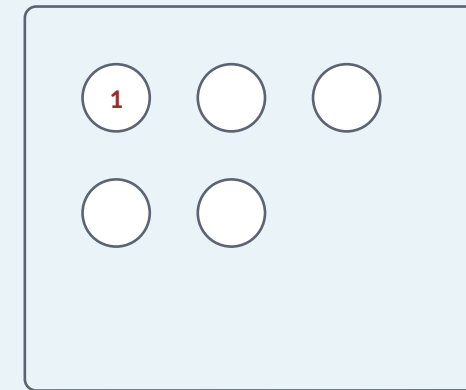
HANDS-ON Deploy the Module 19A workflow to Orchestrator and schedule it.

The same build-then-ship shape as any deployed service, in Orchestrator's own terms

Deployed Still Isn't Automatic

PROBLEM

- A deployed process still only runs when someone starts a job by hand
- That solves "runs outside Studio," not "runs on its own"
- Nobody should have to remember to click Run on the 1st of every month
- The same recurring-schedule need yesterday's own trigger solved, unsolved here



calendar

?

process — idle,
waiting to be started

Deployment answers where the workflow lives, not when it runs

Triggers and Scheduling

SOLUTION

- A trigger "enable[s] you to execute jobs in a preplanned manner"
- Time trigger: runs at regular intervals or a set schedule
- Queue trigger: runs whenever new items are added to a queue
- Meridian Residences' monthly deadline is a time-trigger shape

HANDS-ON Deploy the Module 19A workflow to Orchestrator and schedule it.



TIME TRIGGER

regular interval / schedule



QUEUE TRIGGER

fires on new queue item

TODAY'S CHOICE ↑

Short interval in class — monthly in production

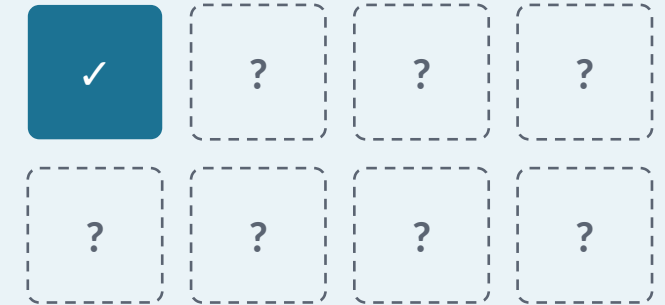
Orchestrator becomes responsible for starting the job on time, every time

One Run Doesn't Prove Every Lease Was Handled

PROBLEM

- A single scheduled run answers "did the process run today"
- It doesn't answer "did all forty active leases get their invoice"
- The workflow as built does one thing, once, against one screen
- Running a separate job per lease wouldn't scale or track status

process ran ✓



8 sample leases — only 1 confirmed

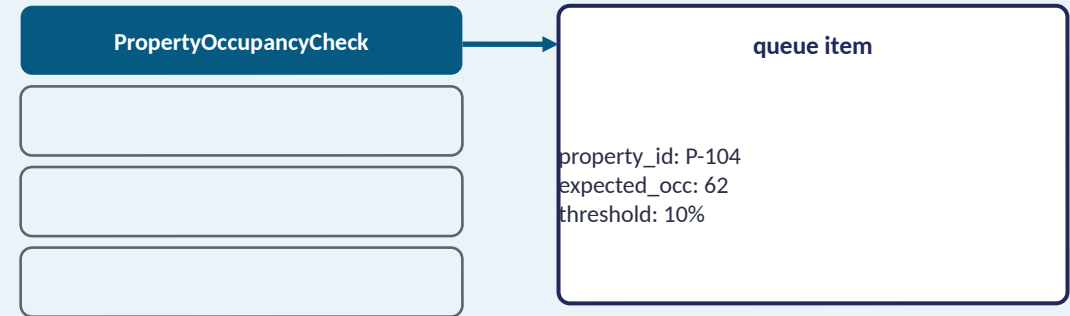
Nothing here tracks which of the other seven ever ran

Queues Hold Repeatable Units of Work

SOLUTION

- A queue: "a container that enables you to hold an unlimited number of items"
- Each queue item carries the data one unit of work needs
- Processed by priority and deadline, otherwise First In, First Out
- A queue item becomes a transaction once a workflow picks it up

HANDS-ON Create a queue and process transactions from it through the deployed workflow.



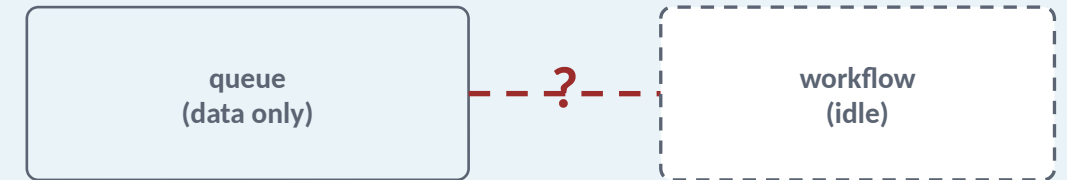
FIFO ↓

Sitting in Orchestrator, waiting to be picked up — not yet a transaction

A Queue Still Needs a Workflow-Side Loop

PROBLEM

- A queue full of items is still just data sitting in Orchestrator
- Something has to ask for the next item, do the work, report back
- Success and failure need to be told apart, not just logged as "done"
- Not every failure deserves the same response



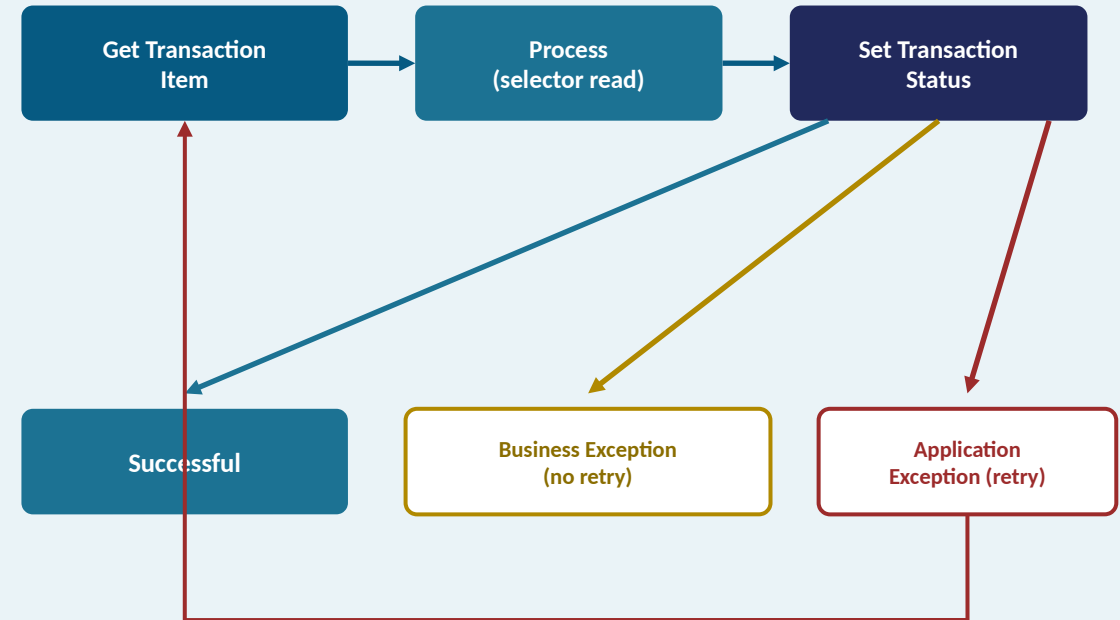
no connection: take one, process it, report back

Orchestrator holds the items — the workflow still has to do the loop

Get Transaction Item, Process, Set Status

SOLUTION

- Get Transaction Item retrieves the next item, sets it In Progress
- The workflow's own logic processes it (today: the Module 19A selector read)
- Set Transaction Status marks Successful or Failed when done
- Failed splits into Business Exception (won't succeed) or Application Exception (retry)



Auto Retry, up to configured limit

One transaction succeeds; one retries; a business failure is logged, never retried

HANDS-ON Create a queue and process transactions from it through the deployed workflow.

Applying This to Your Team's Real Sprint-3 Task

APPLY

- This publish-schedule-queue-transaction shape is a template, not the final answer
- A UiPath team maps it onto its own real screen, queue design, and failure cases
- Each team decides its own success / business-exception / application-exception split
- A Power Automate team treats today as a skills exercise for its already-built deliverable

HANDS-ON For a team choosing UiPath as its Sprint 3 automated workflow, apply this to that capstone task.

TODAY'S GENERIC EXAMPLE

property occupancy
queue item

selector read +
status set

success / exception?

same shape, real data — placeholder fields to fill in per team

YOUR TEAM'S REAL AUTOMATION

Resorts & Spa — own screen,
queue & failure cases

Residences — own screen,
queue & failure cases

Two teams complete their actual Sprint 3 deliverable here; three practiced this shape yesterday